

AUGSBURG, Germany

WMO: 108520

Lat: 48.4253N

Lon: 10.9419E

Elev: 462

StdP: 95.90

Time Zone: 1.00 (EUC)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.3	-10.1	-16.2	1.0	-12.1	-13.1	1.3	-9.3	13.0	6.0	11.4	5.0	2.2	60	0.543

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	30.0	19.8	28.0	18.8	26.1	18.1	20.6	27.5	19.7	26.0	18.9	24.6	2.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.9	22.7	17.5	13.3	21.9	16.8	12.7	21.2	61.3	27.6	58.2	26.2	55.5	24.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 10.0	8.4	7.2	DB	-16.2	32.7	5.2	1.6	-19.9	33.8	-23.0	34.8	-25.9	35.7	-29.7	36.8	(4)
(5)			WB	-16.4	22.2	5.2	0.8	-20.2	22.8	-23.2	23.3	-26.1	23.8	-29.9	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.1	-0.3	4.4	8.4	12.8	16.2	18.1	18.4	13.9	9.4	3.8	1.3	(6)
(7)		DBStd	7.72	4.57	5.42	4.25	3.93	3.70	3.65	2.85	3.31	3.14	3.57	3.39	4.33	(7)
(8)		HDD10.0	1382	306	289	177	74	18	2	0	0	3	53	188	270	(8)
(9)		HDD18.3	3534	564	523	433	297	174	83	41	41	138	276	436	528	(9)
(10)		CDD10.0	990	0	0	3	27	104	188	249	262	119	36	1	1	(10)
(11)		CDD18.3	101	0	0	0	0	2	19	32	44	4	0	0	0	(11)
(12)		CDH23.3	1139	0	0	0	12	27	214	354	477	54	1	0	0	(12)
(13)		CDH26.7	286	0	0	0	1	1	49	85	144	6	0	0	0	(13)
(14)	Wind	WSAvg	2.8	3.5	3.0	3.5	3.0	2.8	2.5	2.4	2.3	2.3	2.6	2.7	3.4	(14)
(15)	Precipitation	PrecAvg	810	48	43	44	60	86	105	92	95	64	56	60	51	(15)
(16)		PrecMax	1055	132	140	114	114	192	235	182	186	130	165	162	114	(16)
(17)		PrecMin	650	13	6	9	21	15	47	30	25	9	3	14	10	(17)
(18)		PrecStd	110	26	25	25	29	42	40	40	37	32	40	34	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	15.7	19.7	25.0	26.2	32.1	32.2	32.9	27.9	22.7	14.9	12.7	(19)
(20)			MCWB	8.4	8.6	11.2	14.8	17.1	20.8	20.2	20.4	18.6	15.3	10.8	9.8	(20)
(21)		2%	DB	9.2	11.3	16.1	21.2	23.9	28.1	29.1	30.2	25.1	20.1	12.0	10.8	(21)
(22)			MCWB	6.6	7.5	9.4	12.8	15.4	19.1	19.1	19.8	17.2	14.5	9.4	8.3	(22)
(23)		5%	DB	7.5	8.8	13.3	18.9	22.2	26.0	27.1	28.1	22.7	17.5	10.2	9.0	(23)
(24)			MCWB	5.6	6.3	8.4	11.8	14.3	18.2	18.5	19.1	16.6	13.2	8.3	7.1	(24)
(25)		10%	DB	6.0	6.2	10.8	16.0	20.6	24.0	25.2	26.1	20.3	15.1	8.9	7.1	(25)
(26)			MCWB	4.5	4.3	7.4	10.5	13.6	17.4	17.8	18.4	15.5	12.4	7.5	5.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.2	11.8	15.5	18.4	22.1	21.2	22.1	19.5	16.2	11.4	10.1	(27)
(28)			MCDWB	11.1	13.4	18.7	23.6	24.7	29.6	28.1	29.2	25.3	21.1	13.1	12.0	(28)
(29)		2%	WB	7.1	7.8	10.0	13.6	16.7	20.1	20.1	20.7	18.2	14.7	9.8	8.6	(29)
(30)			MCDWB	8.8	11.0	14.4	19.9	21.8	26.3	26.9	27.7	23.4	18.6	11.6	10.4	(30)
(31)		5%	WB	5.7	6.3	8.8	12.3	15.3	19.1	19.3	19.7	17.2	13.7	8.7	7.2	(31)
(32)			MCDWB	7.0	8.5	12.5	17.7	20.3	24.6	25.4	26.1	21.4	16.9	10.1	8.8	(32)
(33)		10%	WB	4.6	4.5	7.8	10.9	14.1	17.9	18.5	18.8	16.1	12.5	7.5	5.7	(33)
(34)			MCDWB	5.8	5.9	10.8	15.2	19.1	23.0	23.8	24.7	19.4	14.9	8.7	6.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.3	7.3	8.5	9.9	10.8	10.3	11.0	11.6	10.3	8.6	5.8	5.1	(35)
(36)			MCDBR	7.3	11.5	14.1	15.7	14.0	14.6	15.1	15.8	14.4	13.2	8.7	7.7	(36)
(37)			MCWBR	5.4	8.0	8.3	8.3	6.6	7.0	6.6	6.5	7.5	8.2	6.2	5.8	(37)
(38)		5% WB	MCDBR	6.7	10.9	12.0	14.4	12.6	13.4	13.1	14.0	11.9	12.3	8.1	6.9	(38)
(39)			MCWBR	5.3	7.9	7.6	7.9	6.4	6.8	6.1	6.5	6.5	8.2	6.1	5.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.323	0.357	0.389	0.398	0.403	0.406	0.401	0.377	0.350	0.320	0.295	(40)	
(41)		taud	2.453	2.386	2.335	2.271	2.277	2.296	2.280	2.316	2.362	2.430	2.469	2.460	(41)	
(42)		Ebn at Noon	775	830	855	859	864	860	850	838	823	788	740	732	(42)	
(43)		Edh at Noon	64	85	105	124	128	126	127	117	102	81	63	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.00	1.86	3.01	4.31	5.02	5.63	5.38	4.70	3.38	2.09	1.10	0.80	(44)	
(45)		RadStd	0.14	0.25	0.38	0.57	0.57	0.47	0.48	0.46	0.41	0.25	0.14	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (70 neighbors)		N/A	N/A	N/A	N/A	+0.34	+0.39	N/A	N/A	+59	+20		

Nomenclature: See separate page